# KALEB SOFER

MACHINE LEARNING ENGINEER

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Senior engineer with six years' experience designing and delivering scalable, production-grade data systems, spanning the full lifecycle from infrastructure and ETL to user-facing ML and visualisation. Consulting for both start-ups and established businesses, I have a strong track record of leading multidisciplinary teams.

## TECHNICAL SKILLS

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| **Languages & Engineering** | Python, SQL, TypeScript, Swift, Kotlin, Node.js, Jest, Maestro, Bash, Linux, Git, GitHub Actions, Azure DevOps |
| **ML Frameworks** | PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, NumPy, pandas, spaCy, NetworkX, GraphFrames |
| **LLMs & Retrieval** | Multiple APIs and agent SDKs, MCP, LangGraph, Hugging Face, Sentence Transformers, OpenSearch, Qdrant, FAISS, pgvector, BM25 |
| **Analytics & Visualisation** | Matplotlib, Plotly, Power BI, PostHog, Streamlit |
| **Data Pipelines & Orchestration** | Spark, Kafka, Snowflake, Databricks, BigQuery, PostgreSQL, Hive, dbt, Delta Lake, Apache Iceberg, Great Expectations, Splink, Airflow, Prefect, Kubeflow |
| **Platform, Training & Serving** | AWS, GCP, Azure, Terraform, Docker, Kubernetes, PyTorch DDP/FSDP, FastAPI, ONNX Runtime, MLflow, Weights & Biases, Sentry, Render, Cloudflare Workers, LogRocket |
| **Mobile** | React Native, Expo, EAS Build, HealthKit, Health Connect, Firebase Auth |

## EXPERIENCE

### SOFTMAX LTD — Founding Director
Jun 2024 – Present

Lead client acquisition, discovery and technical delivery for a data and AI consultancy. Our current client is Founders & Coders C.I.C, find us at softmaxco.io

### FOUNDERS & CODERS C.I.C — Senior ML Engineer via Softmax
Jan 2026 – Present

Built FaC's search and knowledge assistant for clients, staff and learners: hybrid BM25/embedding retrieval, cross-encoder re-ranking and cited RAG answers, evaluated on Recall@K, NDCG and LLM-judged faithfulness. Design and deliver Level 6 ML workshops for **up to 40 experienced engineers**: transformers, CNNs, RNNs, distributed and mixed-precision training, quantisation, distillation, dataset versioning and model registries.

**STACK** FastAPI, OpenSearch, pgvector, Claude API, Hugging Face, Sentence Transformers, PyTorch, MLflow, Weights & Biases, ONNX Runtime

### TRACTION — Founding Engineer
Jul 2025 – Present

Took a habit-tracking app from concept to public launch on iOS and Android, solo: native health-data sync, widgets, background tasks and push reminders on a token-authenticated REST backend with rate limiting and moderation. Two-tower recommendation model for social goals; offline-first state across PostgreSQL and on-device stores. Shipped via a custom agentic coding harness with MCP-driven emulator testing, simulated AI users for behaviour and load testing, and CI with over-the-air updates. Grew to **100+ daily signups** through A/B tests, cohort analysis and causal inference on retention. Still growing.

**STACK** TypeScript, React Native, Expo, Swift, Kotlin, HealthKit, Health Connect, Node.js, Express, PostgreSQL, Firebase Auth, PyTorch, Claude API, Claude Agent SDK, MCP, EAS Build, Cloudflare Workers

### MADETECH
Mar 2022 – Apr 2024

#### Ministry of Justice — Senior Data Engineer
Jul 2023 – Apr 2024

Led data engineering for a **live migration of 7,000–10,000 court cases a day across 47 police forces**: CDC-synced Iceberg lakehouse, dual-run reconciliation for zero-downtime cutover, probabilistic record linkage and APIs for front-end services, with automated testing, validation and CI/CD across every pipeline.

**STACK** AWS (S3, Glue, Athena, EMR, Lambda, API Gateway), Apache Iceberg, PySpark, Splink, Great Expectations, Airflow

#### DVLA — Senior Engineer
Mar 2022 – Jun 2023

Led a large team building a **near real-time analytics platform for 40 million UK registered vehicles**: medallion lakehouse, batch and streaming pipelines, and governed data products with full lineage feeding dashboards. Profiled and optimised Spark workloads to cut runtime and compute cost.

**STACK** Databricks, Delta Lake, PySpark, Azure (ADLS, Event Hubs, Data Factory), Azure DevOps, Power BI

### KUBRICK
Mar 2019 – Nov 2021

#### Santander — Data Scientist, Financial Crime
Aug 2020 – Nov 2021

Built the ML for an **anti-money laundering platform used daily by 400 analysts**: entity resolution, engineered transaction features, graph ML, anomaly detection and NER background checks, cutting reviews from half a day to an hour and **saving £12m**. Calibrated supervised models with tuned hyperparameters and leakage-safe cross-validation (PR-AUC, ROC-AUC), with drift monitoring and automated retraining.

**STACK** PySpark, Hive, PyTorch, scikit-learn, XGBoost, spaCy, NetworkX, GraphFrames, MLflow

#### Deliveroo — Data Engineer, Payments
Jun 2019 – Jun 2020

Built and ran a **payments platform giving 115,000 restaurants faster payouts** during COVID-19 lockdowns: double-entry ledger, dimensional models with SCD history, CDC-fed idempotent pipelines, data-quality checks and automated reconciliation.

**STACK** Snowflake, dbt, PostgreSQL, Airflow, AWS (S3, ECS, Lambda, SQS)

## EDUCATION

**ML for experienced software engineers (Apprenticeship Lvl 6)** Founders and Coders — Jan 2026 – Present

**Advanced training in deep learning** The Machine Learning Institute — Sep – Dec 2024

**B.Eng. Aerospace Engineering** University of Manchester — First Class Honours

## INTERESTS

Saxophonist and music producer; qualified sailing skipper, keen swimmer, injured footballer.